

Theoretical Foundations of RADAR-SWARM

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1 Introduction

1.1 Overview of RADAR-SWARM

RADAR-SWARM is a research-oriented simulation framework designed to model and analyze a distributed autonomous airspace monitoring and interception system. The system combines advanced concepts from radar signal processing, estimation theory, multi-agent systems, and control theory to create a unified platform for studying cooperative sensing and decision-making.

The core idea is to deploy a network of radar-equipped agents (either ground-based sensors or aerial drones) that collaboratively detect, track, and intercept multiple dynamic targets in a three-dimensional (3D) environment.

Unlike traditional centralized radar systems, RADAR-SWARM emphasizes:

- **Decentralization:** Each agent operates with partial information.
- **Scalability:** The system can handle increasing numbers of targets and agents.
- **Robustness:** The system tolerates noise, uncertainty, and partial failures.
- **Autonomy:** Agents make local decisions while contributing to global objectives.

1.2 Problem Statement

Modern airspace environments are increasingly complex due to the proliferation of:

- Unmanned Aerial Vehicles (UAVs)
- Low-observable targets
- High-speed maneuvering objects

Traditional radar systems face limitations such as:

- Limited field of view
- Vulnerability to occlusion
- Centralized processing bottlenecks

The primary problem addressed by RADAR-SWARM is:

How can a distributed network of radar-equipped agents collaboratively detect, track, and intercept multiple targets in real time under uncertainty and limited communication?

This problem involves several interconnected challenges:

- Accurate estimation under noisy measurements
- Real-time multi-target tracking
- Efficient sensor fusion across distributed nodes
- Coordination of multiple autonomous agents
- Optimal interception strategies

1.3 System Architecture Overview

The RADAR-SWARM system can be decomposed into several interacting subsystems, each corresponding to a major theoretical domain.

1.3.1 1. Sensing Layer (Radar System)

Each agent is equipped with an FMCW radar system that provides measurements such as:

- Range
- Relative velocity (Doppler)
- Angle of arrival

Mathematically, the measurement process can be expressed as:

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k \quad (1)$$

where:

- $\mathbf{x}_k$ is the true state of the target
- $h(\cdot)$ is the nonlinear measurement function
- $\mathbf{v}_k$ represents measurement noise

1.3.2 2. Signal Processing Layer

Raw radar signals are processed to extract meaningful features:

- Range estimation via beat frequency
- Velocity estimation via Doppler shift
- Angle estimation using antenna arrays

This layer transforms continuous-time signals into discrete measurements usable by tracking algorithms.

1.3.3 3. Estimation and Tracking Layer

The system maintains estimates of target states using probabilistic filters such as:

- Kalman Filter (KF)
- Extended Kalman Filter (EKF)
- Interacting Multiple Model (IMM)

The state evolution is modeled as:

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + \mathbf{w}_k \quad (2)$$

where $\mathbf{w}_k$ is process noise.

1.3.4 4. Sensor Fusion Layer

Multiple agents observe the same targets from different perspectives. These observations are fused to improve accuracy.

Fusion methods include:

- Weighted averaging
- Covariance intersection
- Distributed consensus algorithms

1.3.5 5. Swarm Coordination Layer

Agents coordinate their actions using decentralized control laws. Each agent updates its behavior based on local observations and neighboring agents.

A general control law can be expressed as:

$$\mathbf{u}_i = f(\mathbf{x}_i, \mathbf{x}_{\mathcal{N}_i}) \quad (3)$$

where $\mathcal{N}_i$ represents neighboring agents.

1.3.6 6. Guidance and Interception Layer

Once targets are tracked, interceptor drones are guided using control strategies such as:

- Pursuit guidance
- Proportional navigation

These strategies aim to minimize interception time and energy.

1.3.7 7. Temporal Coordination Layer

All agents must operate under synchronized time to ensure consistency in measurements and decisions.

Synchronization can be modeled as:

$$t_i \rightarrow t_{\text{global}} \quad (4)$$

Advanced concepts such as time-crystal-inspired synchronization may provide robust periodic coordination mechanisms.

1.4 Interdisciplinary Nature of the System

RADAR-SWARM lies at the intersection of multiple disciplines:

- **Signal Processing** — extraction of information from radar signals
- **Control Theory** — guidance and interception
- **Estimation Theory** — tracking under uncertainty
- **Robotics** — swarm coordination
- **Applied Mathematics** — modeling and optimization

1.5 Connection to the Simulator

Each theoretical component in this document directly maps to modules in the RADAR-SWARM simulator:

- Mathematical models $\rightarrow$ state representations in code
- Radar equations $\rightarrow$ signal generation and processing modules
- Tracking algorithms $\rightarrow$ estimation pipelines
- Swarm models $\rightarrow$ agent behaviors

Thus, this document serves as the theoretical backbone for implementing and extending the simulator.

2 Mathematical Foundations

This section establishes the mathematical tools required for modeling, estimation, and decision-making in RADAR-SWARM. The framework relies heavily on linear algebra, probability theory, and state-space representations.

2.1 Linear Algebra Foundations

2.1.1 Vector Spaces

A state in RADAR-SWARM is represented as a vector in $\mathbb{R}^n$:

$$\mathbf{x} \in \mathbb{R}^n \quad (5)$$

For a 3D tracking system, a common state vector is:

$$\mathbf{x} = [x \quad y \quad z \quad v_x \quad v_y \quad v_z]^T \quad (6)$$

This vector encodes both position and velocity.

2.1.2 Matrix Transformations

Linear transformations are represented as:

$$\mathbf{y} = A\mathbf{x} \quad (7)$$

where:

- $A \in \mathbb{R}^{m \times n}$
- $\mathbf{x} \in \mathbb{R}^n$
- $\mathbf{y} \in \mathbb{R}^m$

These transformations are used in:

- State propagation
- Measurement projection

2.1.3 Eigenvalues and Eigenvectors

For matrix A :

$$A\mathbf{v} = \lambda\mathbf{v} \quad (8)$$

Eigenvalues determine system stability and are crucial in:

- Dynamical systems
- Observability and controllability

2.1.4 Norms and Distance

Euclidean norm:

$$\|\mathbf{x}\| = \sqrt{x_1^2 + x_2^2 + \dots + x_n^2} \quad (9)$$

Distance between two states:

$$d = \|\mathbf{x}_1 - \mathbf{x}_2\| \quad (10)$$

Used in:

- Target association
- Clustering

2.1.5 Covariance Matrices

Uncertainty is represented using covariance:

$$P = E[(\mathbf{x} - \mu)(\mathbf{x} - \mu)^T] \quad (11)$$

Properties:

- Symmetric: $P = P^T$
- Positive semi-definite

Interpretation:

- Diagonal elements $\rightarrow$ variance
- Off-diagonal elements $\rightarrow$ correlation

2.2 Probability Theory

2.2.1 Random Variables

A random variable X maps outcomes to real values:

$$X : \Omega \rightarrow \mathbb{R} \quad (12)$$

2.2.2 Gaussian Distribution

Most noise in radar systems is modeled as Gaussian:

$$p(x) = \frac{1}{\sqrt{2\pi\sigma^2}} \exp\left(-\frac{(x - \mu)^2}{2\sigma^2}\right) \quad (13)$$

Multivariate Gaussian:

$$p(\mathbf{x}) = \frac{1}{(2\pi)^{n/2} |P|^{1/2}} \exp\left(-\frac{1}{2}(\mathbf{x} - \mu)^T P^{-1}(\mathbf{x} - \mu)\right) \quad (14)$$

2.2.3 Expectation and Variance

Expectation:

$$E[X] = \int xp(x)dx \quad (15)$$

Variance:

$$\text{Var}(X) = E[(X - \mu)^2] \quad (16)$$

2.2.4 Conditional Probability

$$P(A|B) = \frac{P(A \cap B)}{P(B)} \quad (17)$$

This is fundamental for Bayesian estimation.

2.2.5 Bayesian Inference

Posterior update:

$$p(\mathbf{x}|\mathbf{z}) \propto p(\mathbf{z}|\mathbf{x})p(\mathbf{x}) \quad (18)$$

Used in:

- Kalman filtering
- Sensor fusion

2.2.6 Noise Models in RADAR-SWARM

Two key noise sources:

Process Noise:

$$\mathbf{w}_k \sim \mathcal{N}(0, Q) \quad (19)$$

Measurement Noise:

$$\mathbf{v}_k \sim \mathcal{N}(0, R) \quad (20)$$

Where:

- Q = process covariance
- R = measurement covariance

2.3 State-Space Models

The RADAR-SWARM system is modeled as a discrete-time dynamical system.

2.3.1 State Transition Model

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + B\mathbf{u}_k + \mathbf{w}_k \quad (21)$$

where:

- F = state transition matrix
- B = control input matrix
- $\mathbf{u}_k$ = control input

2.3.2 Measurement Model

$$\mathbf{z}_k = H\mathbf{x}_k + \mathbf{v}_k \quad (22)$$

where:

- H = observation matrix

2.3.3 Nonlinear Models

In radar systems, measurements are often nonlinear:

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k \quad (23)$$

Example:

$$r = \sqrt{x^2 + y^2 + z^2} \quad (24)$$

$$\theta = \tan^{-1}(y/x) \quad (25)$$

2.3.4 Observability

A system is observable if the state can be reconstructed from measurements.

Observability matrix:

$$\mathcal{O} = \begin{bmatrix} H \\ HF \\ HF^2 \\ \vdots \end{bmatrix} \quad (26)$$

Full rank $\Rightarrow$ system is observable.

2.3.5 Controllability

A system is controllable if it can be driven to any state.

$$\mathcal{C} = [B \quad FB \quad F^2B \quad \dots] \quad (27)$$

Full rank $\Rightarrow$ controllable.

2.4 Connection to RADAR-SWARM

These mathematical tools enable:

- Representation of targets as state vectors
- Modeling uncertainty using covariance matrices
- Predicting motion using linear systems
- Updating estimates using probabilistic inference

Without this foundation, advanced components like Kalman filters, sensor fusion, and swarm coordination cannot be rigorously implemented.

3 3D Kinematics and Motion Models

In RADAR-SWARM, accurate modeling of target and agent motion is essential for prediction, tracking, and interception. This section develops continuous-time and discrete-time motion models used in estimation algorithms.

3.1 Continuous-Time Kinematics

3.1.1 Position, Velocity, and Acceleration

The motion of a target in 3D space is described by:

$$\mathbf{p}(t) = \begin{bmatrix} x(t) \\ y(t) \\ z(t) \end{bmatrix} \quad (28)$$

$$\mathbf{v}(t) = \frac{d\mathbf{p}(t)}{dt} \quad (29)$$

$$\mathbf{a}(t) = \frac{d\mathbf{v}(t)}{dt} \quad (30)$$

Thus:

$$\frac{d^2\mathbf{p}(t)}{dt^2} = \mathbf{a}(t) \quad (31)$$

3.1.2 Newtonian Motion Model

Assuming constant acceleration:

$$\mathbf{p}(t) = \mathbf{p}_0 + \mathbf{v}_0 t + \frac{1}{2} \mathbf{a} t^2 \quad (32)$$

$$\mathbf{v}(t) = \mathbf{v}_0 + \mathbf{a} t \quad (33)$$

These equations form the basis for discrete-time models.

3.2 Discrete-Time Modeling

In simulation, time is discretized:

$$t_k = k\Delta t \quad (34)$$

3.2.1 State Vector Definition

For tracking:

$$\mathbf{x}_k = [x \quad y \quad z \quad v_x \quad v_y \quad v_z]^T \quad (35)$$

3.3 Constant Velocity (CV) Model

Assumes acceleration is zero or negligible.

3.3.1 State Transition Equation

$$\mathbf{x}_{k+1} = F\mathbf{x}_k + \mathbf{w}_k \quad (36)$$

where:

$$F = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (37)$$

3.3.2 Interpretation

- Position evolves using velocity
- Velocity remains constant

3.3.3 Process Noise

To account for unmodeled acceleration:

$$\mathbf{w}_k \sim \mathcal{N}(0, Q) \quad (38)$$

A common process noise covariance:

$$Q = q \begin{bmatrix} \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} & 0 & 0 \\ 0 & \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} & 0 \\ 0 & 0 & \frac{\Delta t^4}{4} & 0 & 0 & \frac{\Delta t^3}{2} \\ \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 & 0 & 0 \\ 0 & \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 & 0 \\ 0 & 0 & \frac{\Delta t^3}{2} & 0 & 0 & \Delta t^2 \end{bmatrix} \quad (39)$$

3.4 Constant Acceleration (CA) Model

More accurate for maneuvering targets.

3.4.1 State Vector

$$\mathbf{x}_k = [x \quad y \quad z \quad v_x \quad v_y \quad v_z \quad a_x \quad a_y \quad a_z]^T \quad (40)$$

3.4.2 State Transition Matrix

$$F = \begin{bmatrix} 1 & 0 & 0 & \Delta t & 0 & 0 & \frac{1}{2}\Delta t^2 & 0 & 0 \\ 0 & 1 & 0 & 0 & \Delta t & 0 & 0 & \frac{1}{2}\Delta t^2 & 0 \\ 0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 & \frac{1}{2}\Delta t^2 \\ 0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t & 0 \\ 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & \Delta t \\ 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \quad (41)$$

3.5 Nearly Constant Turn Model (Advanced)

For maneuvering targets such as drones or aircraft.

Angular velocity ω :

$$\theta_{k+1} = \theta_k + \omega \Delta t \quad (42)$$

Velocity rotates:

$$\begin{bmatrix} v_x \\ v_y \end{bmatrix} = \begin{bmatrix} \cos(\omega \Delta t) & -\sin(\omega \Delta t) \\ \sin(\omega \Delta t) & \cos(\omega \Delta t) \end{bmatrix} \begin{bmatrix} v_x \\ v_y \end{bmatrix} \quad (43)$$

3.6 Relative Motion (Radar Perspective)

Radar measures relative motion:

$$\mathbf{r} = \mathbf{p}_{target} - \mathbf{p}_{sensor} \quad (44)$$

Relative velocity:

$$\mathbf{v}_{rel} = \mathbf{v}_{target} - \mathbf{v}_{sensor} \quad (45)$$

This is crucial for Doppler estimation.

3.7 Discrete-Time Integration

Using Euler approximation:

$$\mathbf{p}_{k+1} = \mathbf{p}_k + \mathbf{v}_k \Delta t \quad (46)$$

$$\mathbf{v}_{k+1} = \mathbf{v}_k + \mathbf{a}_k \Delta t \quad (47)$$

Higher-order methods (Runge-Kutta) can improve accuracy but increase computational cost.

3.8 Uncertainty Propagation

State covariance evolves as:

$$P_{k+1} = F P_k F^T + Q \quad (48)$$

This equation is fundamental for Kalman filtering.

3.9 Connection to RADAR-SWARM

These motion models are used in:

- **Target Simulation:** Generating realistic trajectories
- **Prediction Step:** Kalman filter state propagation
- **Interception:** Predicting future positions

Model selection depends on target behavior:

- CV $\rightarrow$ simple motion
- CA $\rightarrow$ accelerating targets
- Turn model $\rightarrow$ maneuvering drones

Choosing the correct model significantly impacts tracking accuracy and interception success.

4 FMCW Radar Theory

Frequency-Modulated Continuous Wave (FMCW) radar is the primary sensing mechanism in RADAR-SWARM. It enables simultaneous estimation of target range and velocity using frequency-modulated signals.

4.1 Basic Principle of FMCW Radar

Unlike pulsed radar, FMCW radar transmits a continuous signal whose frequency varies over time (chirp signal). The key idea is:

- Transmit a known frequency-modulated signal
- Receive a delayed and Doppler-shifted echo
- Mix transmitted and received signals to obtain beat frequency

This beat frequency encodes both range and velocity information.

4.2 Chirp Signal Model

The transmitted FMCW signal is a linear frequency-modulated (LFM) chirp:

$$s_t(t) = A \cos \left(2\pi \left(f_c t + \frac{S t^2}{2} \right) \right) \quad (49)$$

where:

- A = amplitude
- f_c = carrier frequency
- $S = \frac{B}{T}$ = chirp slope
- B = bandwidth
- T = chirp duration

Instantaneous frequency:

$$f(t) = f_c + St \quad (50)$$

4.3 Received Signal Model

A target at distance R introduces a time delay:

$$\tau = \frac{2R}{c} \quad (51)$$

The received signal:

$$s_r(t) = A \cos \left(2\pi \left(f_c(t - \tau) + \frac{S(t - \tau)^2}{2} \right) \right) \quad (52)$$

4.4 Beat Frequency Derivation

The transmitted and received signals are mixed:

$$s_b(t) = s_t(t) \cdot s_r(t) \quad (53)$$

After low-pass filtering, the beat frequency is:

$$f_b = S\tau \quad (54)$$

Substituting τ :

$$f_b = \frac{2SR}{c} \quad (55)$$

4.5 Range Estimation

Solving for range:

$$R = \frac{cf_b}{2S} \quad (56)$$

4.6 Doppler Effect

If the target is moving with radial velocity v , the received signal experiences a Doppler shift:

$$f_D = \frac{2vf_c}{c} \quad (57)$$

4.7 Combined Range and Doppler

In practice, the beat frequency contains both range and Doppler components:

$$f_b = \frac{2SR}{c} + f_D \quad (58)$$

This creates ambiguity, which is resolved using multiple chirps.

4.8 Multiple Chirp Processing

By transmitting a sequence of chirps:

- Range is estimated from fast-time (within chirp)
- Velocity is estimated from slow-time (across chirps)

Phase difference between chirps:

$$\Delta\phi = 2\pi f_D T \quad (59)$$

Velocity:

$$v = \frac{\lambda\Delta\phi}{4\pi T} \quad (60)$$

where $\lambda = \frac{c}{f_c}$.

4.9 Range Resolution

$$\Delta R = \frac{c}{2B} \quad (61)$$

Higher bandwidth $\Rightarrow$ better resolution.

4.10 Velocity Resolution

$$\Delta v = \frac{\lambda}{2NT} \quad (62)$$

where N is number of chirps.

4.11 Maximum Unambiguous Range

$$R_{max} = \frac{cT}{2} \quad (63)$$

4.12 Maximum Unambiguous Velocity

$$v_{max} = \frac{\lambda}{4T} \quad (64)$$

4.13 Radar Equation (Power Model)

Received power:

$$P_r = \frac{P_t G^2 \lambda^2 \sigma}{(4\pi)^3 R^4} \quad (65)$$

where:

- P_t = transmitted power
- G = antenna gain
- σ = radar cross-section

4.14 Noise in Radar Systems

Thermal noise power:

$$P_n = kTB \quad (66)$$

Signal-to-noise ratio (SNR):

$$\text{SNR} = \frac{P_r}{P_n} \quad (67)$$

4.15 Practical Considerations

- Chirp linearity affects accuracy
- Phase noise introduces errors
- Multipath reflections distort signals

4.16 Connection to RADAR-SWARM

In the simulator:

- Chirp signals are synthetically generated
- Beat frequencies are computed from target states
- Range and velocity are extracted from simulated signals

Each radar-equipped agent independently performs these operations, forming the sensing backbone of the swarm system.

5 Radar Signal Processing

Radar signal processing converts raw time-domain signals into meaningful target information such as range and velocity. In FMCW radar systems, this is achieved using Fourier-based methods and structured signal processing pipelines.

5.1 Signal Representation

After mixing transmitted and received signals, the beat signal can be represented in discrete-time as:

$$x[n, m] \quad (68)$$

where:

- n = fast-time index (within a chirp)
- m = slow-time index (across chirps)

Thus, the radar data forms a 2D matrix:

$$X \in \mathbb{C}^{N \times M} \quad (69)$$

5.2 Range Processing (Fast-Time FFT)

Range estimation is performed by applying a Fourier transform along the fast-time dimension.

5.2.1 Discrete Fourier Transform (DFT)

$$X[k] = \sum_{n=0}^{N-1} x[n] e^{-j2\pi \frac{kn}{N}} \quad (70)$$

5.2.2 Interpretation

- Frequency bins correspond to beat frequencies
- Beat frequency $\rightarrow$ target range

5.2.3 Range Mapping

Using:

$$R = \frac{cf_b}{2S} \quad (71)$$

Each FFT bin maps to a specific range.

5.3 Doppler Processing (Slow-Time FFT)

To estimate velocity, a second FFT is applied across chirps.

5.3.1 Doppler FFT

$$X_d[k] = \sum_{m=0}^{M-1} x[m] e^{-j2\pi \frac{km}{M}} \quad (72)$$

5.3.2 Interpretation

- Phase changes across chirps encode velocity
- Doppler frequency $\rightarrow$ radial velocity

5.4 Range-Doppler Map

Combining both FFT operations yields a 2D representation:

$$RDM(k_r, k_d) = |FFT_{slow}(FFT_{fast}(x[n, m]))|^2 \quad (73)$$

5.4.1 Interpretation

- Horizontal axis: range bins
- Vertical axis: Doppler bins
- Peaks indicate detected targets

5.5 Resolution and Sampling

5.5.1 Range Resolution

$$\Delta R = \frac{c}{2B} \quad (74)$$

5.5.2 Velocity Resolution

$$\Delta v = \frac{\lambda}{2MT} \quad (75)$$

5.5.3 Sampling Requirements

Nyquist condition:

$$f_s \geq 2f_{max} \quad (76)$$

5.6 Windowing Effects

To reduce spectral leakage, window functions are applied before FFT.

5.6.1 Window Function

$$x_w[n] = x[n] \cdot w[n] \quad (77)$$

Common windows:

- Hamming
- Hanning
- Blackman

5.6.2 Trade-Off

- Reduced sidelobes
- Increased main-lobe width (lower resolution)

5.7 Zero Padding

Improves interpolation of FFT results:

$$x'[n] = \begin{cases} x[n], & n < N \\ 0, & n \geq N \end{cases} \quad (78)$$

5.8 Clutter and Noise Suppression

Static objects create clutter near zero Doppler.

5.8.1 High-Pass Filtering

Removes low-frequency components:

$$y[n] = x[n] - x[n - 1] \quad (79)$$

5.9 Peak Detection

Targets are identified as peaks in the Range-Doppler Map.

$$(k_r^*, k_d^*) = \arg \max RDM(k_r, k_d) \quad (80)$$

5.10 Computational Complexity

FFT complexity:

$$\mathcal{O}(N \log N) \quad (81)$$

Total processing:

- Range FFT: $N \log N$
- Doppler FFT: $M \log M$

5.11 Connection to RADAR-SWARM

In the simulator:

- Raw signals are generated using FMCW equations
- Fast-time FFT extracts range
- Slow-time FFT extracts velocity
- Range-Doppler maps provide detections

Each radar node independently performs this pipeline, producing measurements that are later used for tracking and fusion.

This stage acts as the bridge between physical signal generation and higher-level estimation algorithms.

6 Detection Theory

Detection theory provides the mathematical framework for distinguishing target signals from noise in radar systems. In RADAR-SWARM, this stage converts Range-Doppler maps into reliable target detections.

6.1 Signal Detection Problem

The detection problem is formulated as a binary hypothesis test:

$$H_0 : \text{Noise only (no target)} \quad (82)$$

$$H_1 : \text{Signal + Noise (target present)} \quad (83)$$

Observed signal:

$$z = \begin{cases} n, & H_0 \\ s + n, & H_1 \end{cases} \quad (84)$$

where:

- s = target signal
- n = noise

6.2 Noise Modeling

Radar noise is typically modeled as Gaussian:

$$n \sim \mathcal{N}(0, \sigma^2) \quad (85)$$

In power domain (after FFT), noise follows an exponential distribution:

$$p(z) = \frac{1}{\sigma^2} e^{-z/\sigma^2} \quad (86)$$

6.3 Threshold Detection

A detection is declared if:

$$z > T \quad (87)$$

where T is a threshold.

6.4 Performance Metrics

6.4.1 Probability of Detection

$$P_D = P(z > T | H_1) \quad (88)$$

6.4.2 Probability of False Alarm

$$P_{FA} = P(z > T | H_0) \quad (89)$$

6.4.3 Trade-Off

- High threshold $\Rightarrow$ fewer false alarms but missed targets
- Low threshold $\Rightarrow$ more detections but higher false alarms

6.5 Constant False Alarm Rate (CFAR)

CFAR adaptively adjusts the threshold based on local noise estimation.

6.5.1 Basic Idea

- Estimate noise from neighboring cells
- Set threshold relative to noise level

6.5.2 CFAR Threshold

$$T = \alpha \cdot \hat{N} \quad (90)$$

where:

- $\hat{N}$ = estimated noise power
- α = scaling factor

6.6 Cell Structure in CFAR

- **Cell Under Test (CUT)**
- **Guard Cells** (exclude target leakage)
- **Training Cells** (estimate noise)

6.7 Cell Averaging CFAR (CA-CFAR)

Noise estimate:

$$\hat{N} = \frac{1}{N} \sum_{i=1}^N z_i \quad (91)$$

Threshold:

$$T = \alpha \cdot \hat{N} \quad (92)$$

6.8 Ordered Statistic CFAR (OS-CFAR)

Sort training cells:

$$z_{(1)} \leq z_{(2)} \leq \dots \leq z_{(N)} \quad (93)$$

Select k -th value:

$$\hat{N} = z_{(k)} \quad (94)$$

More robust against:

- Clutter
- Multiple targets

6.9 2D CFAR in Range-Doppler Map

CFAR is applied over the 2D RDM:

$$T(r, d) = \alpha \cdot \text{mean of surrounding cells} \quad (95)$$

Detection:

$$\text{Detect if } RDM(r, d) > T(r, d) \quad (96)$$

6.10 Sliding Window Implementation

For each cell:

1. Define window
2. Exclude guard cells
3. Compute noise estimate
4. Apply threshold

6.11 SNR-Based Detection

Signal-to-noise ratio:

$$\text{SNR} = \frac{S}{N} \quad (97)$$

Detection improves as SNR increases.

6.12 Clutter Effects

Clutter includes:

- Ground reflections
- Buildings
- Static objects

CFAR helps mitigate clutter influence.

6.13 Detection Output

After CFAR, detections are represented as:

$$\mathcal{D} = \{(r_i, d_i, A_i)\} \quad (98)$$

where:

- r_i = range
- d_i = Doppler
- A_i = amplitude

6.14 Connection to RADAR-SWARM

In the simulator:

- Range-Doppler maps are computed
- CFAR is applied to extract detections
- Output detections become inputs to tracking algorithms

This stage is critical because:

- Poor detection $\Rightarrow$ bad tracking
- Good detection $\Rightarrow$ stable multi-target tracking

Thus, detection theory acts as the gateway between signal processing and intelligent decision-making in RADAR-SWARM.

7 Angle of Arrival Estimation

Angle of Arrival (AoA) estimation determines the direction from which a signal is received. In RADAR-SWARM, this enables full 3D localization of targets when combined with range and Doppler measurements.

7.1 Antenna Array Model

AoA estimation requires multiple antennas arranged in a known geometry. A common configuration is the Uniform Linear Array (ULA).

7.1.1 Uniform Linear Array (ULA)

Assume:

- M antenna elements
- Inter-element spacing d
- Incoming plane wave with angle θ

The signal arrives at each antenna with a phase shift.

7.2 Time Delay and Phase Shift

The path difference between adjacent elements is:

$$\Delta l = d \sin \theta \quad (99)$$

Corresponding time delay:

$$\Delta t = \frac{d \sin \theta}{c} \quad (100)$$

Phase shift:

$$\Delta \phi = \frac{2\pi}{\lambda} d \sin \theta \quad (101)$$

where $\lambda = \frac{c}{f_c}$.

7.3 Steering Vector

The received signal across the array can be represented using a steering vector:

$$\mathbf{a}(\theta) = \begin{bmatrix} 1 \\ e^{-jkd \sin \theta} \\ e^{-j2kd \sin \theta} \\ \vdots \\ e^{-j(M-1)kd \sin \theta} \end{bmatrix} \quad (102)$$

where $k = \frac{2\pi}{\lambda}$ is the wave number.

7.4 Signal Model

For K incoming signals:

$$\mathbf{x}(t) = A\mathbf{s}(t) + \mathbf{n}(t) \quad (103)$$

where:

- $A = [\mathbf{a}(\theta_1), \dots, \mathbf{a}(\theta_K)]$
- $\mathbf{s}(t)$ = signal vector
- $\mathbf{n}(t)$ = noise

7.5 Spatial Covariance Matrix

$$R = E[\mathbf{x}\mathbf{x}^H] \quad (104)$$

Expanding:

$$R = AR_s A^H + \sigma^2 I \quad (105)$$

where:

- R_s = signal covariance
- $\sigma^2 I$ = noise covariance

7.6 Eigenvalue Decomposition

$$R = E\Lambda E^H \quad (106)$$

Eigenvectors are divided into:

- Signal subspace
- Noise subspace

7.7 MUSIC Algorithm

MUSIC (Multiple Signal Classification) exploits the orthogonality between signal and noise subspaces.

7.7.1 Key Idea

$$\mathbf{a}^H(\theta)E_n = 0 \quad (107)$$

where E_n is the noise subspace.

7.7.2 MUSIC Spectrum

$$P(\theta) = \frac{1}{\mathbf{a}^H(\theta)E_n E_n^H \mathbf{a}(\theta)} \quad (108)$$

7.7.3 Angle Estimation

$$\theta_i = \arg \max P(\theta) \quad (109)$$

Peaks in the MUSIC spectrum correspond to target directions.

7.8 Resolution and Limitations

- Higher number of antennas $\Rightarrow$ better resolution
- Requires uncorrelated signals
- Sensitive to noise and calibration errors

7.9 2D Angle Estimation (Azimuth and Elevation)

Using planar arrays, both azimuth (θ) and elevation (ϕ) can be estimated:

$$\mathbf{a}(\theta, \phi) \quad (110)$$

This enables full 3D localization.

7.10 Beamforming (Alternative Method)

Simpler method:

$$P(\theta) = |\mathbf{w}^H \mathbf{x}|^2 \quad (111)$$

Less accurate than MUSIC but computationally cheaper.

7.11 Connection to RADAR-SWARM

In RADAR-SWARM:

- Each radar node uses an antenna array
- AoA estimation provides direction of targets
- Combined with range:

$$\text{Target Position} \Rightarrow (R, \theta, \phi) \quad (112)$$

This enables:

- 3D tracking
- Multi-sensor triangulation
- Improved sensor fusion accuracy

Without AoA estimation, the system cannot perform accurate spatial localization, making this component essential for the RADAR-SWARM architecture.

8 Measurement Models

Measurement models define how radar observations relate to the true state of a target. In RADAR-SWARM, sensors provide measurements in polar coordinates, which must be transformed into Cartesian space for tracking and control.

8.1 State Representation

The target state is represented in Cartesian coordinates:

$$\mathbf{x} = [x \quad y \quad z \quad v_x \quad v_y \quad v_z]^T \quad (113)$$

8.2 Radar Measurement Vector

A typical radar measurement consists of:

$$\mathbf{z} = \begin{bmatrix} r \\ \theta \\ \phi \\ \dot{r} \end{bmatrix} \quad (114)$$

where:

- r = range
- θ = azimuth angle
- ϕ = elevation angle
- $\dot{r}$ = radial velocity (Doppler)

8.3 Nonlinear Measurement Function

The relationship between state and measurement is nonlinear:

8.3.1 Range

$$r = \sqrt{x^2 + y^2 + z^2} \quad (115)$$

8.3.2 Azimuth Angle

$$\theta = \tan^{-1} \left(\frac{y}{x} \right) \quad (116)$$

8.3.3 Elevation Angle

$$\phi = \tan^{-1} \left(\frac{z}{\sqrt{x^2 + y^2}} \right) \quad (117)$$

8.3.4 Radial Velocity

$$\dot{r} = \frac{xv_x + yv_y + zv_z}{\sqrt{x^2 + y^2 + z^2}} \quad (118)$$

8.4 Compact Measurement Model

$$\mathbf{z}_k = h(\mathbf{x}_k) + \mathbf{v}_k \quad (119)$$

where:

- $h(\cdot)$ is a nonlinear function
- $\mathbf{v}_k \sim \mathcal{N}(0, R)$ is measurement noise

8.5 Conversion to Cartesian Coordinates

Sometimes measurements are converted to Cartesian form:

$$x = r \cos \phi \cos \theta \quad (120)$$

$$y = r \cos \phi \sin \theta \quad (121)$$

$$z = r \sin \phi \quad (122)$$

However, this transformation introduces correlated noise.

8.6 Measurement Noise Characteristics

Measurement noise is typically modeled as:

$$R = \begin{bmatrix} \sigma_r^2 & 0 & 0 & 0 \\ 0 & \sigma_\theta^2 & 0 & 0 \\ 0 & 0 & \sigma_\phi^2 & 0 \\ 0 & 0 & 0 & \sigma_r^2 \end{bmatrix} \quad (123)$$

Key insight:

- Range noise is usually small
- Angle noise significantly affects position accuracy at long distances

8.7 Jacobian Matrix for EKF

Because the measurement model is nonlinear, the Extended Kalman Filter requires linearization.

8.7.1 Jacobian Definition

$$H = \frac{\partial h(\mathbf{x})}{\partial \mathbf{x}} \quad (124)$$

8.7.2 Partial Derivatives for Range

$$\frac{\partial r}{\partial x} = \frac{x}{r}, \quad \frac{\partial r}{\partial y} = \frac{y}{r}, \quad \frac{\partial r}{\partial z} = \frac{z}{r} \quad (125)$$

8.7.3 Partial Derivatives for Azimuth

$$\frac{\partial \theta}{\partial x} = -\frac{y}{x^2 + y^2}, \quad \frac{\partial \theta}{\partial y} = \frac{x}{x^2 + y^2} \quad (126)$$

8.7.4 Partial Derivatives for Elevation

$$\frac{\partial \phi}{\partial z} = \frac{\sqrt{x^2 + y^2}}{r^2} \quad (127)$$

8.8 Measurement Linearization

The nonlinear model is approximated as:

$$\mathbf{z}_k \approx h(\hat{\mathbf{x}}_k) + H_k(\mathbf{x}_k - \hat{\mathbf{x}}_k) \quad (128)$$

This enables EKF updates.

8.9 Multiple Sensor Measurements

For multiple radar nodes:

$$\mathbf{z}_k^{(i)} = h^{(i)}(\mathbf{x}_k) + \mathbf{v}_k^{(i)} \quad (129)$$

Each sensor has:

- Different position
- Different noise characteristics

8.10 Observability Considerations

Single radar limitations:

- Poor depth resolution with angle noise

Multiple radars:

- Improve observability
- Enable triangulation

8.11 Connection to RADAR-SWARM

This model is used in:

- EKF measurement update
- Multi-sensor fusion
- Target localization

Pipeline:

- Radar outputs: $(r, \theta, \phi, \dot{r})$
- Measurement model maps to state space
- EKF updates state estimate

This section is essential because it links raw radar observations to the mathematical framework used for tracking and prediction in RADAR-SWARM.

9 Sensor Fusion

Sensor fusion combines information from multiple sensors to produce a more accurate and reliable estimate of the system state. In RADAR-SWARM, multiple radar-equipped agents observe the same targets from different perspectives, enabling improved accuracy, robustness, and observability.

9.1 Motivation for Sensor Fusion

Single-sensor limitations:

- Limited field of view
- Measurement noise
- Occlusion and shadowing

Advantages of multi-sensor fusion:

- Reduced uncertainty
- Improved accuracy
- Increased robustness to failures

9.2 Measurement Fusion Model

Assume N sensors observing the same target:

$$\mathbf{z}_i = H_i \mathbf{x} + \mathbf{v}_i \tag{130}$$

where:

- $\mathbf{z}_i$ = measurement from sensor i
- H_i = measurement matrix
- $\mathbf{v}_i \sim \mathcal{N}(0, R_i)$

9.3 Centralized Fusion

All measurements are combined into a single update:

$$\mathbf{z} = \begin{bmatrix} \mathbf{z}_1 \\ \mathbf{z}_2 \\ \vdots \\ \mathbf{z}_N \end{bmatrix} \quad (131)$$

$$H = \begin{bmatrix} H_1 \\ H_2 \\ \vdots \\ H_N \end{bmatrix} \quad (132)$$

Then standard Kalman update is applied.

9.4 Weighted Least Squares Fusion

If multiple estimates $\hat{\mathbf{x}}_i$ with covariance P_i are available:

$$\hat{\mathbf{x}} = \left(\sum_{i=1}^N P_i^{-1} \right)^{-1} \left(\sum_{i=1}^N P_i^{-1} \hat{\mathbf{x}}_i \right) \quad (133)$$

9.4.1 Interpretation

- Sensors with lower uncertainty (smaller P_i) have higher weight

9.5 Covariance Intersection (CI)

When cross-correlation between estimates is unknown:

$$P^{-1} = \omega P_1^{-1} + (1 - \omega) P_2^{-1} \quad (134)$$

$$\hat{\mathbf{x}} = P \left(\omega P_1^{-1} \hat{\mathbf{x}}_1 + (1 - \omega) P_2^{-1} \hat{\mathbf{x}}_2 \right) \quad (135)$$

where:

- $\omega \in [0, 1]$

9.5.1 Key Advantage

- Avoids overconfidence due to unknown correlations

9.6 Distributed Sensor Fusion

In RADAR-SWARM, fusion is decentralized.

Each agent:

- Maintains its own estimate
- Exchanges information with neighbors

9.6.1 Consensus Algorithm

$$\mathbf{x}_i^{(k+1)} = \mathbf{x}_i^{(k)} + \sum_{j \in \mathcal{N}_i} w_{ij} (\mathbf{x}_j^{(k)} - \mathbf{x}_i^{(k)}) \quad (136)$$

where:

- $\mathcal{N}_i$ = neighbors of agent i
- w_{ij} = weighting factor

9.7 Information Filter Form

Instead of covariance form, use information matrix:

$$Y = P^{-1}, \quad \mathbf{y} = Y\mathbf{x} \quad (137)$$

Fusion becomes additive:

$$Y = \sum Y_i \quad (138)$$

$$\mathbf{y} = \sum \mathbf{y}_i \quad (139)$$

9.8 Track-to-Track Fusion

Each agent runs its own tracker and shares estimates.

Fusion:

$$\hat{\mathbf{x}} = f(\hat{\mathbf{x}}_1, \hat{\mathbf{x}}_2, \dots) \quad (140)$$

Challenges:

- Correlated estimates
- Communication delays

9.9 Geometric Fusion (Triangulation)

Multiple sensors observing same target:

$$\mathbf{p} = \arg \min \sum_i \|\mathbf{z}_i - h_i(\mathbf{p})\|^2 \quad (141)$$

Used for:

- Position refinement
- Reducing angle uncertainty

9.10 Fusion Under Communication Constraints

In real systems:

- Limited bandwidth
- Delayed updates

Solutions:

- Event-triggered communication
- Compressed state representation

9.11 Robust Fusion

Outlier rejection:

$$\text{Reject if } (\mathbf{z} - H\hat{\mathbf{x}})^T S^{-1} (\mathbf{z} - H\hat{\mathbf{x}}) > \gamma \quad (142)$$

9.12 Connection to RADAR-SWARM

Sensor fusion enables:

- Multiple radar nodes tracking the same target
- Improved accuracy over single-node tracking
- Robust operation under partial failure

Pipeline:

1. Each radar detects and tracks targets locally
2. Estimates are shared across the network
3. Fusion algorithms combine information
4. Global consistent state is obtained

This transforms RADAR-SWARM into a distributed intelligence system capable of collaborative sensing and decision-making.

10 Swarm Robotics

Swarm robotics studies the collective behavior of multiple autonomous agents that interact locally to achieve global objectives. In RADAR-SWARM, interceptor drones and radar nodes operate as a decentralized multi-agent system.

10.1 Multi-Agent System Model

Consider a swarm of N agents:

$$\mathcal{A} = \{1, 2, \dots, N\} \quad (143)$$

Each agent has a state:

$$\mathbf{x}_i = \begin{bmatrix} \mathbf{p}_i \\ \mathbf{v}_i \end{bmatrix} \quad (144)$$

and control input:

$$\mathbf{u}_i \quad (145)$$

Dynamics:

$$\mathbf{x}_i(k+1) = f(\mathbf{x}_i(k), \mathbf{u}_i(k)) \quad (146)$$

10.2 Decentralized Control

Each agent makes decisions based on:

- Its own state
- Information from neighbors

Control law:

$$\mathbf{u}_i = f(\mathbf{x}_i, \mathbf{x}_{\mathcal{N}_i}) \quad (147)$$

where $\mathcal{N}_i$ denotes neighboring agents.

10.3 Communication Graph

The swarm is modeled as a graph:

$$\mathcal{G} = (\mathcal{V}, \mathcal{E}) \quad (148)$$

where:

- $\mathcal{V}$ = agents
- $\mathcal{E}$ = communication links

Adjacency matrix:

$$A_{ij} = \begin{cases} 1, & \text{if agent } j \text{ communicates with } i \\ 0, & \text{otherwise} \end{cases} \quad (149)$$

10.4 Consensus Algorithms

Agents aim to agree on a common value.

$$\mathbf{x}_i(k+1) = \mathbf{x}_i(k) + \sum_{j \in \mathcal{N}_i} w_{ij}(\mathbf{x}_j(k) - \mathbf{x}_i(k)) \quad (150)$$

10.4.1 Interpretation

- Each agent adjusts toward neighbors
- Global agreement emerges from local interactions

10.5 Formation Control

Agents maintain a desired formation.

$$\mathbf{u}_i = - \sum_{j \in \mathcal{N}_i} (\mathbf{p}_i - \mathbf{p}_j - \mathbf{d}_{ij}) \quad (151)$$

where $\mathbf{d}_{ij}$ is desired relative position.

10.6 Flocking Behavior (Reynolds Rules)

Three fundamental rules:

- **Separation:** avoid collisions
- **Alignment:** match velocity
- **Cohesion:** stay close to group

Combined control:

$$\mathbf{u}_i = \mathbf{u}_{sep} + \mathbf{u}_{align} + \mathbf{u}_{coh} \quad (152)$$

10.7 Task Allocation

Assign tasks (e.g., intercept target) to agents.

10.7.1 Optimization-Based Assignment

$$\min \sum_{i,j} C_{ij} x_{ij} \quad (153)$$

where:

- C_{ij} = cost (distance, energy)
- $x_{ij} \in \{0, 1\}$

10.7.2 Market-Based Methods

Agents bid for tasks:

- Lower cost $\Rightarrow$ higher priority

10.8 Coverage Control

Agents distribute to maximize sensing coverage.

$$\mathbf{p}_i = \arg \min \int_{\mathcal{V}_i} \|\mathbf{p} - \mathbf{p}_i\|^2 d\mathbf{p} \quad (154)$$

10.9 Collision Avoidance

Maintain safe distance:

$$\|\mathbf{p}_i - \mathbf{p}_j\| > d_{min} \quad (155)$$

Repulsive force:

$$\mathbf{u}_{rep} \propto \frac{\mathbf{p}_i - \mathbf{p}_j}{\|\mathbf{p}_i - \mathbf{p}_j\|^3} \quad (156)$$

10.10 Decentralized Decision Making

Each agent evaluates:

- Target priority
- Distance to target
- Available resources

Decision rule:

$$\text{Select target} = \arg \min_j C_{ij} \quad (157)$$

10.11 Emergent Behavior

Global behavior emerges from local rules:

- Coordinated tracking
- Efficient interception
- Adaptive coverage

10.12 Scalability

Swarm systems scale well because:

- No central controller
- Local communication only

10.13 Connection to RADAR-SWARM

In RADAR-SWARM:

- Radar nodes share information via a communication graph
- Interceptor drones coordinate to intercept targets
- Task allocation assigns targets to drones

Pipeline:

1. Targets detected and tracked
2. Swarm evaluates priorities
3. Agents coordinate via consensus
4. Tasks assigned and executed

This enables fully autonomous, decentralized operation, making the system robust and scalable in complex environments.

11 Guidance and Interception

Guidance and interception theory governs how an interceptor agent moves to intercept a target based on estimated target states. In RADAR-SWARM, this is the final stage where sensing, tracking, and decision-making translate into physical action.

11.1 Interception Problem Formulation

Let:

- $\mathbf{p}_t$ = target position
- $\mathbf{p}_i$ = interceptor position

Relative position:

$$\mathbf{r} = \mathbf{p}_t - \mathbf{p}_i \quad (158)$$

Relative velocity:

$$\mathbf{v}_r = \mathbf{v}_t - \mathbf{v}_i \quad (159)$$

Goal:

$$\mathbf{r}(t_f) = 0 \quad (160)$$

11.2 Line-of-Sight (LOS) Geometry

The line-of-sight angle λ is defined as:

$$\lambda = \tan^{-1} \left(\frac{y_t - y_i}{x_t - x_i} \right) \quad (161)$$

LOS rate:

$$\dot{\lambda} = \frac{d\lambda}{dt} \quad (162)$$

11.3 Pure Pursuit Guidance

The interceptor always points toward the target:

$$\mathbf{u} \propto \mathbf{r} \quad (163)$$

11.3.1 Characteristics

- Simple implementation
- Inefficient for fast targets

11.4 Proportional Navigation (PN)

Proportional Navigation is the most widely used interception strategy.

11.4.1 Control Law

$$a_n = N v_c \dot{\lambda} \quad (164)$$

where:

- a_n = normal acceleration
- N = navigation constant (typically 3–5)
- v_c = closing velocity

11.4.2 Closing Velocity

$$v_c = -\frac{d}{dt} \|\mathbf{r}\| \quad (165)$$

11.4.3 Interpretation

- Interceptor reacts to changes in LOS angle
- Stabilizes interception trajectory

11.5 Vector Form of PN

$$\mathbf{a} = N(\mathbf{v}_r \times \hat{\mathbf{r}}) \times \hat{\mathbf{r}} \quad (166)$$

where:

- $\hat{\mathbf{r}} = \frac{\mathbf{r}}{\|\mathbf{r}\|}$

11.6 Time-to-Intercept Estimation

Assuming constant velocities:

$$t_f = \frac{\|\mathbf{r}\|}{\|\mathbf{v}_r\|} \quad (167)$$

More accurate estimation solves:

$$\mathbf{p}_t + \mathbf{v}_i t = \mathbf{p}_i + \mathbf{v}_i t \quad (168)$$

11.7 Lead Pursuit Strategy

Instead of chasing the current position, aim at predicted position:

$$\mathbf{p}_{aim} = \mathbf{p}_t + \mathbf{v}_t t_f \quad (169)$$

11.8 Interception Constraints

- Maximum acceleration
- Velocity limits
- Energy constraints

11.9 Optimal Control Perspective

Minimize cost function:

$$J = \int_0^{t_f} (\|\mathbf{u}(t)\|^2 + \alpha t) dt \quad (170)$$

11.10 Multi-Agent Interception

Multiple interceptors coordinate:

- Assign different targets
- Surround target
- Minimize interception time

11.11 Cooperative Interception

Agents share target information:

$$\mathbf{u}_i = f(\mathbf{x}_i, \mathbf{x}_t, \mathbf{x}_{neighbors}) \quad (171)$$

11.12 Collision Avoidance During Interception

Constraint:

$$\|\mathbf{p}_i - \mathbf{p}_j\| > d_{safe} \quad (172)$$

11.13 Robust Interception Under Uncertainty

Target state is estimated:

$$\hat{\mathbf{x}}_t, \quad P_t \quad (173)$$

Guidance must account for uncertainty:

- Use predicted trajectories
- Add safety margins

11.14 Connection to RADAR-SWARM

In RADAR-SWARM:

- Target states are estimated via tracking
- Swarm assigns interceptors to targets
- Guidance laws compute control inputs
- Agents move toward interception points

Pipeline:

1. Target detected and tracked
2. Future trajectory predicted
3. Interceptor assigned
4. Guidance law applied

This section represents the final step in the perception-to-action pipeline, enabling RADAR-SWARM to actively respond to dynamic threats.

12 Adaptive Radar Geometry

Adaptive radar geometry studies how the spatial arrangement of sensors affects the accuracy of target localization. In RADAR-SWARM, multiple radar nodes and drones dynamically adjust their positions to improve estimation performance.

12.1 Motivation

Localization accuracy depends not only on measurement quality but also on sensor geometry.

Poor geometry leads to:

- High uncertainty
- Ambiguous position estimates

Good geometry leads to:

- Accurate triangulation
- Reduced estimation error

12.2 Triangulation Principle

Assume multiple sensors observing a target:

$$\mathbf{z}_i = h_i(\mathbf{p}) + \mathbf{v}_i \quad (174)$$

Estimate target position:

$$\hat{\mathbf{p}} = \arg \min \sum_i \|\mathbf{z}_i - h_i(\mathbf{p})\|^2 \quad (175)$$

12.3 Linearized Measurement Model

Linearizing around $\hat{\mathbf{p}}$:

$$\mathbf{z} \approx H\mathbf{p} + \mathbf{v} \quad (176)$$

where H is the geometry (Jacobian) matrix.

12.4 Error Covariance of Position Estimate

$$P = (H^T R^{-1} H)^{-1} \quad (177)$$

This matrix determines estimation uncertainty.

12.5 Geometric Dilution of Precision (GDOP)

GDOP quantifies the effect of geometry on estimation accuracy:

$$GDOP = \sqrt{\text{trace}(P)} \quad (178)$$

12.5.1 Interpretation

- Low GDOP $\Rightarrow$ good geometry
- High GDOP $\Rightarrow$ poor geometry

12.6 Intuition Behind GDOP

- Sensors close together $\Rightarrow$ poor triangulation
- Sensors widely separated $\Rightarrow$ better accuracy

12.7 Optimal Sensor Placement

Goal:

$$\min_{\{\mathbf{p}_i\}} GDOP \quad (179)$$

subject to:

- Communication constraints
- Physical limits

12.8 Fisher Information Matrix (FIM)

$$J = H^T R^{-1} H \quad (180)$$

Relation:

$$P = J^{-1} \quad (181)$$

Maximizing J improves estimation accuracy.

12.9 D-Optimal Design

Maximize determinant of FIM:

$$\max \det(J) \quad (182)$$

This minimizes uncertainty volume.

12.10 Adaptive Positioning Strategy

Each agent updates position:

$$\mathbf{u}_i = -\nabla_{\mathbf{p}_i} GDOP \quad (183)$$

Agents move to improve geometry.

12.11 Baseline Distance Considerations

- Too small $\Rightarrow$ poor angular resolution
- Too large $\Rightarrow$ communication issues

12.12 Angle Diversity

Sensors should observe targets from different angles:

- Improves triangulation
- Reduces ambiguity

12.13 Dynamic Geometry in RADAR-SWARM

Unlike static systems, RADAR-SWARM allows:

- Moving sensors (drones)
- Adaptive reconfiguration

12.14 Multi-Target Geometry Optimization

For multiple targets:

$$\min \sum_j GDOP_j \quad (184)$$

Trade-off:

- Focus on high-priority targets
- Balance coverage

12.15 Geometric Constraints

- Terrain limitations
- Flight constraints
- Communication range

12.16 Connection to RADAR-SWARM

Adaptive geometry enables:

- Improved tracking accuracy
- Better sensor fusion
- Efficient swarm deployment

Pipeline:

1. Estimate target positions
2. Compute GDOP
3. Adjust sensor positions
4. Improve measurement quality

This allows RADAR-SWARM to actively optimize its sensing configuration, making it a self-improving intelligent system.

13 Temporal Coordination Systems

Temporal coordination ensures that all agents in RADAR-SWARM operate under a consistent notion of time. Accurate synchronization is critical for coherent sensing, data fusion, and coordinated action.

13.1 Importance of Time Synchronization

In distributed systems, each agent maintains its own local clock:

$$t_i = t + \delta_i \tag{185}$$

where:

- t = global time
- δ_i = clock offset of agent i

Without synchronization:

- Measurements become inconsistent
- Sensor fusion produces errors
- Tracking degrades

13.2 Clock Models

Each local clock can be modeled as:

$$t_i(t) = \alpha_i t + \beta_i \tag{186}$$

where:

- α_i = clock drift (frequency offset)
- β_i = clock offset

13.3 Synchronization Objective

Goal:

$$t_i \rightarrow t_j \quad \forall i, j \quad (187)$$

i.e., all clocks converge to a common time.

13.4 Pairwise Synchronization

Agents exchange timestamps:

$$t_1 = t + \delta_1 \quad (188)$$

$$t_2 = t + \delta_2 \quad (189)$$

Offset estimation:

$$\hat{\delta} = \frac{(t_2 - t_1)}{2} \quad (190)$$

13.5 Network-Wide Synchronization

Using consensus:

$$t_i^{(k+1)} = t_i^{(k)} + \sum_{j \in \mathcal{N}_i} w_{ij} (t_j^{(k)} - t_i^{(k)}) \quad (191)$$

All clocks converge to a common value.

13.6 Time Synchronization Error

Synchronization error:

$$e_{ij} = t_i - t_j \quad (192)$$

Goal:

$$e_{ij} \rightarrow 0 \quad (193)$$

13.7 Impact on Radar Measurements

Time errors affect:

- Range estimation
- Doppler estimation

Range error:

$$\Delta R = \frac{c\Delta t}{2} \quad (194)$$

Even small timing errors cause significant spatial errors.

13.8 Time-Stamped Measurements

Each measurement is tagged:

$$\mathbf{z}_k^{(i)}(t_i) \quad (195)$$

Fusion requires alignment:

$$\mathbf{z}_k^{(i)}(t_i) \rightarrow \mathbf{z}_k^{(i)}(t) \quad (196)$$

13.9 Asynchronous Sensor Fusion

If synchronization is imperfect:

- Use interpolation
- Use delayed-state Kalman filters

13.10 Event-Based Coordination

Instead of strict time steps:

- Agents act on events (detections)
- Reduces synchronization burden

13.11 Time-Crystal-Inspired Synchronization

A conceptual advanced approach inspired by physics:

- Systems exhibit stable periodic oscillations
- Synchronization emerges naturally

Model:

$$\dot{\theta}_i = \omega_i + \sum_j K_{ij} \sin(\theta_j - \theta_i) \quad (197)$$

This resembles the Kuramoto model.

13.12 Phase Synchronization

Each agent has a phase:

$$\theta_i(t) \quad (198)$$

Synchronization condition:

$$\theta_i - \theta_j \rightarrow 0 \quad (199)$$

13.13 Advantages of Oscillator-Based Synchronization

- Robust to noise
- Decentralized
- Scalable

13.14 Practical Synchronization Methods

- GPS-based synchronization
- Network Time Protocol (NTP)
- Precision Time Protocol (PTP)

13.15 Connection to RADAR-SWARM

Temporal coordination ensures:

- Consistent radar measurements
- Accurate sensor fusion
- Coordinated swarm behavior

Pipeline:

1. Synchronize clocks across agents
2. Timestamp measurements
3. Align data in fusion stage
4. Execute coordinated actions

This layer acts as the invisible backbone of RADAR-SWARM, ensuring that all distributed processes operate coherently in time.

14 Simulation Architecture

The simulation architecture defines how all components of RADAR-SWARM interact in a discrete-time computational framework. It integrates sensing, estimation, fusion, swarm coordination, and control into a unified system.

14.1 Discrete-Time Simulation Framework

The system operates in discrete time steps:

$$t_k = k\Delta t \tag{200}$$

At each time step, the system updates:

- Target states
- Sensor measurements
- State estimates
- Agent actions

14.2 System State Representation

The global state includes:

- Target states: $\{\mathbf{x}_t^{(j)}\}$
- Agent states: $\{\mathbf{x}_i\}$
- Sensor measurements

14.3 Simulation Loop

At each time step k :

1. **Target Propagation**

$$\mathbf{x}_{t,k+1} = F\mathbf{x}_{t,k} + \mathbf{w}_k \quad (201)$$

2. **Sensor Measurement Generation**

$$\mathbf{z}_k^{(i)} = h^{(i)}(\mathbf{x}_{t,k}) + \mathbf{v}_k^{(i)} \quad (202)$$

3. **Radar Signal Processing**

- Generate FMCW signals
- Apply FFT
- Produce Range-Doppler maps

4. **Detection (CFAR)**

- Extract target detections

5. **Tracking Update**

- Prediction step
- Data association
- Update step (KF/EKF/IMM)

6. **Sensor Fusion**

- Combine multi-agent estimates

7. **Swarm Decision Making**

- Task allocation
- Target prioritization

8. **Guidance and Control**

$$\mathbf{u}_i = f(\mathbf{x}_i, \hat{\mathbf{x}}_t) \quad (203)$$

9. **Agent Motion Update**

$$\mathbf{x}_{i,k+1} = f(\mathbf{x}_{i,k}, \mathbf{u}_i) \quad (204)$$

10. **Geometry Optimization**

- Adjust sensor positions based on GDOP

11. **Time Synchronization**

- Align timestamps across agents

14.4 Agent-Based Modeling

Each agent is modeled as an independent entity with:

- State $\mathbf{x}_i$
- Sensor model
- Communication module
- Decision logic

Agent update:

$$\mathbf{x}_i(k+1) = f(\mathbf{x}_i(k), \mathbf{u}_i(k)) \quad (205)$$

14.5 Modular System Design

The simulation is divided into modules:

- **Environment Module:** target dynamics
- **Sensor Module:** radar simulation
- **Processing Module:** FFT and detection
- **Tracking Module:** filters and association
- **Fusion Module:** multi-sensor integration
- **Swarm Module:** coordination and decisions
- **Control Module:** guidance laws

14.6 Data Flow Architecture

$$\text{Targets} \rightarrow \text{Sensors} \rightarrow \text{Processing} \rightarrow \text{Detection} \quad (206)$$

$$\rightarrow \text{Tracking} \rightarrow \text{Fusion} \rightarrow \text{Swarm} \rightarrow \text{Control} \quad (207)$$

14.7 Parallelization and Scalability

The system supports parallel execution:

- Each agent runs independently
- Sensor processing can be parallelized

14.8 Simulation Parameters

Key parameters:

- Time step Δt
- Number of agents
- Number of targets
- Noise covariance (Q, R)

14.9 Performance Metrics

Evaluate system performance using:

- Tracking error:

$$\|\mathbf{x}_{true} - \hat{\mathbf{x}}\| \quad (208)$$

- Detection rate (P_D)
- False alarm rate (P_{FA})
- Interception success rate

14.10 Visualization and Analysis

Simulation outputs include:

- Target trajectories
- Agent trajectories
- Range-Doppler maps
- Tracking estimates

14.11 Connection to RADAR-SWARM Implementation

This architecture maps directly to code:

- Each section corresponds to a module/class
- Discrete loop corresponds to simulation step function
- Agents implemented as objects

14.12 End-to-End Pipeline Summary

1. Targets move in environment
2. Radars generate signals
3. Signals processed into detections
4. Tracking estimates target states
5. Fusion improves accuracy
6. Swarm assigns tasks
7. Interceptors engage targets

14.13 Conclusion

The simulation architecture integrates all theoretical components into a coherent framework. It enables experimentation, validation, and extension of advanced concepts in radar systems, multi-agent coordination, and autonomous interception.

15 References

References

- [1] M. I. Skolnik, *Introduction to Radar Systems*, McGraw-Hill, 2001.
- [2] Y. Bar-Shalom, *Estimation with Applications to Tracking and Navigation*, Wiley, 2001.
- [3] R. Schmidt, "Multiple emitter location and signal parameter estimation," *IEEE Transactions on Antennas and Propagation*, 1986.
- [4] R. Olfati-Saber, "Consensus and cooperation in networked multi-agent systems," *Proceedings of the IEEE*, 2007.
- [5] P. Zarchan, *Tactical and Strategic Missile Guidance*, AIAA, 2012.